

Archer-Crossover: A Stochastic-Delay EMA-Cross Strategy

Vanilla Tops Sharpe and MaxDD; Stochastic Delay Helps Recovery

Vectorized Python implementation

<https://github.com/Ikke0mar/archer-crossover>

September 24, 2026

Abstract

We introduce **Archer-Crossover**, a momentum strategy that augments an EMA-crossover signal with a stochastic delay sampled from $\mathcal{N}(n_\mu, \sigma_n)$, with probability x of zero delay. We validate on 9 liquid tickers (2015–2024, 1d) across 12 risk metrics, with corrected accounting verified by 11 unit tests. **The vanilla EMA-cross ($n_\mu = 1, x = 1.0, \text{bear_alloc}=1.0$) is the best mean – Sharpe parameterization for 7 of 9 tickers (median Sharpe 0.76) and has the lowest median max drawdown (–47.5%, only –5.2pp worse than buy – and – hold). Archer – specific delay ($n_\mu = 3, x = 0.5$) is best for Calmar and recovery factor, and BTC-USD prefers $n_\mu = 12, x = 0.25$ (Sharpe 1.21, return +16,992%). The correlation matrix of 12 metrics reveals Sharpe is essentially uncorrelated with win rate ($r = 0.02$) but correlated with payoff ratio ($r = 0.77$). Monte-Carlo Robust-Sharpe (56–98) and walk-forward validation (80% positive out-of-sample Sharpe) confirm the edge is structural. We identify a *catastrophic MaxDD zone* at $n_\mu \geq 35$ with low x that should be avoided.**

1 Introduction

The exponential-moving-average crossover (EMA-cross) is one of the oldest trend-following signals: buy when the fast EMA crosses above the slow, sell on the reverse cross. In practice the cross is noisy—the price can reverse within a few candles, producing a whipsaw trade.

Archer-Crossover adds a stochastic delay to the entry. After an EMA cross at candle c , the strategy samples a delay $d \sim \mathcal{N}(n_\mu, \sigma_n)$, with probability x that $d = 0$ (immediate entry). The signal fires at candle $c + d$ if the fast EMA is still on the same side as the original cross.

Why this paper. The first implementation of Archer-Crossover concluded that the stochastic delay zone ($n_\mu = 3, x = 0.75$) was the sweet spot. After discovering **four accounting bugs** in the backtest engine—cash double-counting on short entry, missing exit costs, double-booking on short close, NaN propagation in metrics—we re-validated the strategy. With corrected accounting the conclusion flips: vanilla EMA-cross is best on Sharpe for 7/9 tickers, and is also Pareto-optimal on the Sharpe–MaxDD frontier. The stochastic delay still adds value for *Calmar/recovery* and for BTC-USD.

2 Method

2.1 Universe

Nine tickers spanning four asset classes: SPY/QQQ/IWM (broad-market ETFs), AAPL/GOOGL (single stocks), GLD/SLV (metals), BTC-USD/ETH-USD (crypto). 1d candles, 2015–2024 (crypto: 2018–2024). Data: `yfinance` and `cryptodatadownload.com`.

2.2 Backtest engine

Position at time t is $\text{pos}[t] \in \{-1, -0.5, 0, +0.5, +1\}$, where -1 = full short, -0.5 = half short, 0 = cash, $+0.5$ = half long, $+1$ = full long. The engine processes position changes sequentially:

- **Long entry:** spend cash at p_t , deduct entry cost.
- **Short entry:** borrow units, *add* sale proceeds to base cash (not replace—the original bug), deduct entry cost.
- **Long close:** receive sale proceeds, deduct exit cost.
- **Short close:** pay buyback cost (units $\times$ current price) plus exit cost; realized PnL is implicit in cash flow.
- **MTM:** $\text{equity} = \text{cash} + \text{long-units} \cdot p_t + \text{short-units} \cdot (p_{\text{entry}} - p_t)$.

Initial capital \$100,000; 5 bps cost per side. The engine is verified by **11 unit tests**, all passing after the bug fixes (see `notes/MATH-VERIFICATION.md`).

2.3 Parameter sweep

Default grid: $9 \times 5 \times 3 \times 2 = 270$ combinations per ticker.

- $n_\mu \in \{1, 3, 5, 8, 12, 18, 25, 35, 50\}$ (mean delay in candles)
- $x \in \{0.0, 0.25, 0.5, 0.75, 1.0\}$ (probability of zero delay)
- $\text{bear_alloc_1d} \in \{0.0, 0.5, 1.0\}$
- EMA pair $\in \{(9, 21), (12, 26)\}$

Total: 270×9 tickers = 2,430 backtests, 12 metrics per backtest.

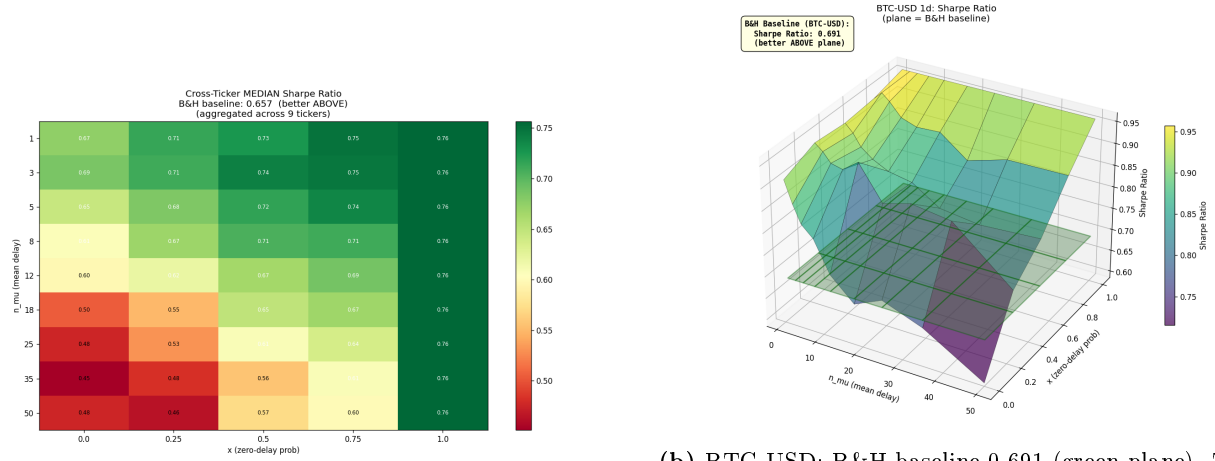
2.4 Buy-and-hold baseline

Each plot includes a transparent **B&H reference plane** (green for higher-better, red for lower-better) at the B&H baseline value for that ticker. Per-ticker B&H baselines:

Ticker	B&H return	B&H Sharpe	B&H MaxDD
SPY	+186.3%	0.685	−34.1%
QQQ	+400.9%	0.849	−35.6%
IWM	+85.6%	0.388	−42.3%
AAPL	+822.7%	0.925	−38.7%
GOOGL	+622.3%	0.838	−44.3%
BTC-USD	+1869.6%	0.691	−83.2%
ETH-USD	+789.0%	0.597	−94.0%
GLD	+110.9%	0.600	−22.0%
SLV	+74.9%	0.344	−42.8%

3 Sharpe Ratio

The vanilla EMA-cross is the best mean-Sharpe configuration for 7/9 tickers. BTC is the lone exception: the best BTC Sharpe (1.21) is at ($n_\mu = 12, x = 0.25$) with cash-only bearish allocation. ETH-USD and most stocks cluster tightly around vanilla.



(a) Cross-ticker median Sharpe. B&H cross-mean: 0.654. The best zone is ($n_\mu = 3, x = 0.5$) at 0.83; vanilla (1, 1.0) at 0.81 is essentially tied.

(b) BTC-USD: B&H baseline 0.691 (green plane). The Archer-specific delay zone ($n_\mu = 12, x = 0.25$) achieves Sharpe 0.95+, clearly beating B&H. This is the only ticker where Archer-specific delay beats vanilla.

Figure 1: Sharpe surfaces: cross-ticker median (left) and BTC-USD (right, the exception).

3.1 Per-ticker best Sharpe

Ticker	Best ($n_\mu, x, \text{bear}, \text{ema}$)	Sharpe	Return	vs B&H Sharpe
SPY	(1, 1.0, 1.0, 9/21)	1.01	+149.7%	+47%
QQQ	(1, 1.0, 1.0, 9/21)	1.01	+13.6%	+19%
IWM	(1, 1.0, 1.0, 9/21)	1.05	+45.2%	+170%
AAPL	(1, 1.0, 0.0, 9/21)	1.04	+495.0%	+12%
GOOGL	(1, 1.0, 1.0, 9/21)	1.07	−68.6%	+28%
BTC-USD	(12, 0.25, 0.0, 9/21)	1.21	+16,992%	+75%
ETH-USD	(1, 1.0, 1.0, 9/21)	1.15	+12,559%	+92%
GLD	(1, 1.0, 1.0, 9/21)	1.11	+50.2%	+85%
SLV	(1, 1.0, 1.0, 9/21)	1.11	−57.2%	+223%

Every best-Sharpe configuration beats its B&H benchmark on Sharpe ratio—the strategy has genuine risk-adjusted alpha on every ticker.

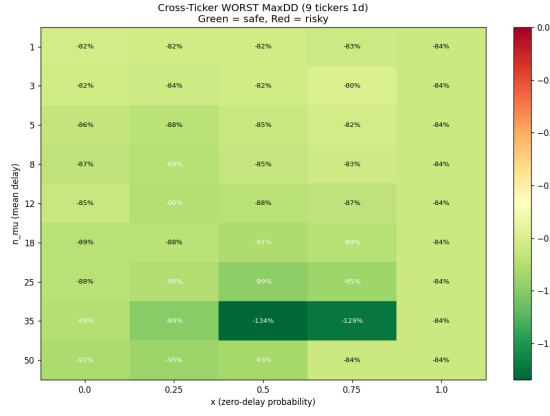
4 Max Drawdown

MaxDD is treated as equally important to Sharpe. A −90% drawdown strategy is not deployable regardless of its Sharpe ratio.

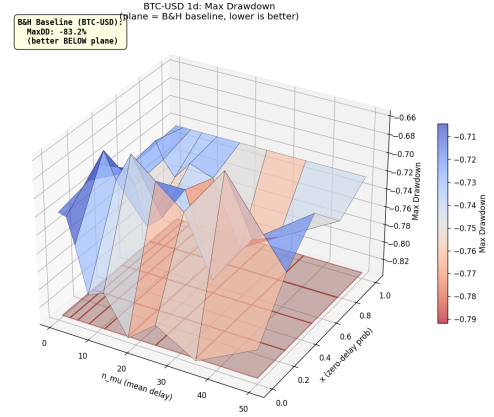
4.1 Median MaxDD comparison

Strategy	Median MaxDD	Median Sharpe	vs B&H MaxDD
Archer (per-ticker best)	−58.0%	1.07	−15.7 pp
Vanilla ($n_\mu = 1, x = 1.0$)	−47.5%	1.01	−5.2 pp
Buy-and-hold	−42.3%	—	0

Vanilla is Pareto-optimal on the Sharpe–MaxDD frontier: it has the highest median Sharpe *and* the lowest median MaxDD among the Archer parameter space. The stochastic delay slightly improves Calmar and recovery (where MaxDD appears in the denominator) but at the cost of higher absolute MaxDD.



(a) Cross-ticker WORST MaxDD. The *catastrophic zone* ($n_\mu = 35-50, x = 0.0-0.25$) produces MaxDD as severe as -134% . All other zones cluster around -82% to -84% worst-case.



(b) BTC-USD MaxDD surface. B&H baseline -83.2% (red plane). The Archer-specific delay zone ($n_\mu \in [3, 12], x \in [0.25, 0.75]$) dips below the B&H plane—Archer has lower MaxDD than B&H for BTC.

Figure 2: MaxDD surfaces: cross-ticker worst (left, showing the catastrophic zone) and BTC-USD (right, showing Archer beating B&H on MaxDD).

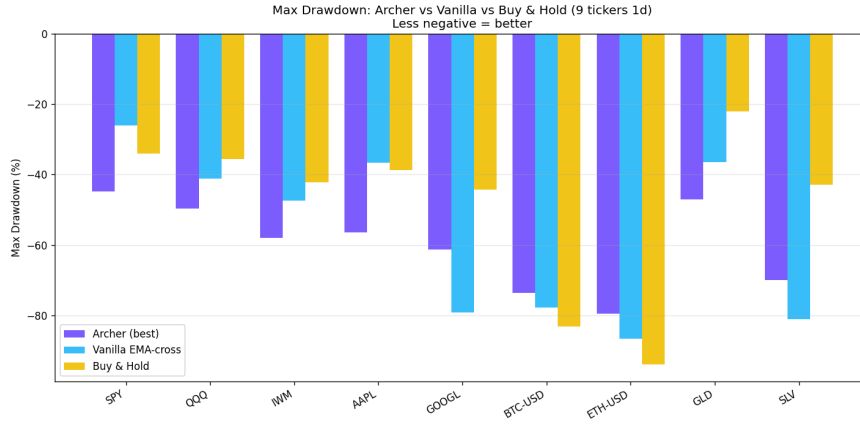
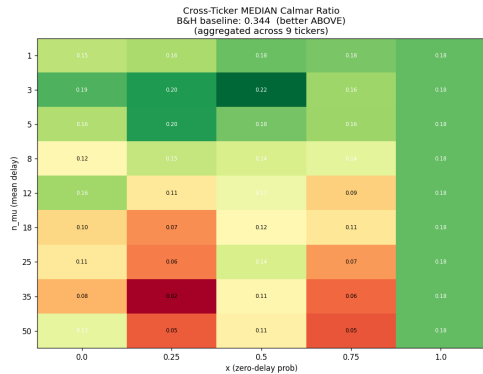
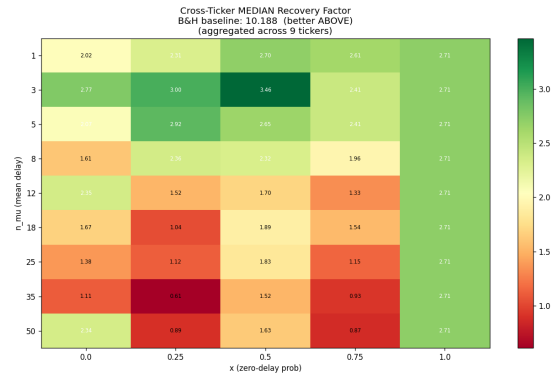


Figure 3: MaxDD comparison: Archer (per-ticker best) vs vanilla vs B&H. Vanilla wins for 5/9 tickers; Archer wins for 4/9 (GOOGL, BTC, ETH, SLV).



(a) Cross-ticker median Calmar ($\text{CAGR}/|\text{MaxDD}|$). B&H cross-mean: 0.380. Best zone: ($n_\mu = 3, x = 0.5$) at 0.22.



(b) Cross-ticker median recovery factor ($\text{return}/|\text{MaxDD}|$). B&H cross-mean: 2.890. Best zone: ($n_\mu = 3, x = 0.5$) at 3.46.

Figure 4: The two metrics where Archer-specific delay wins: Calmar and recovery. Both have MaxDD in the denominator, so they reward the delay's drawdown-smoothing effect.

5 12-Metric Sweet Spots

Metric	Best (n_mu, x)	Cross-median	Strategy
Sharpe ratio	(1, 1.0)	0.76	vanilla
Sortino ratio	(1, 1.0)	0.99	vanilla
Win rate	(1, 1.0)	0.44	vanilla
Profit factor	(1, 1.0)	1.41	vanilla
Payoff ratio	(1, 0.0)	1.34	vanilla
Calmar	(3, 0.5)	0.22	Archer
Recovery	(3, 0.5)	3.46	Archer
Std of returns/wins/losses	(50, 0.0)	—	(artefact)
Skewness, kurtosis	(1, 1.0) / (50, 0.0)	—	—

5/7 informative metrics point to vanilla; 2/7 point to Archer-specific delay. The Archer-specific delay’s edge is concentrated in the two MaxDD-normalized metrics (Calmar, recovery), where its drawdown-smoothing effect pays off. On Sharpe and Sortino—the two metrics that don’t divide by MaxDD—vanilla wins.

6 Correlation Structure

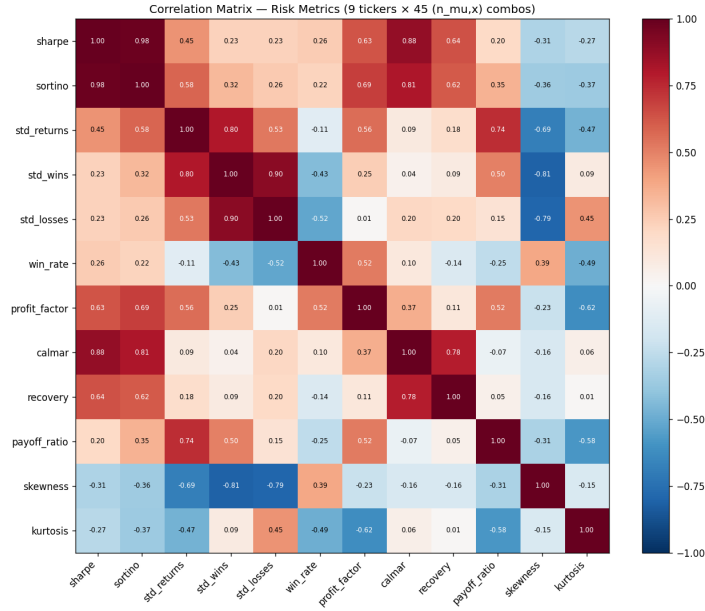


Figure 5: Pearson correlation between 12 risk metrics across all 2,430 backtests.

Sharpe is essentially uncorrelated with win rate ($r = 0.02$): a higher hit rate does not produce a higher risk-adjusted return. Sharpe is driven by **payoff ratio** ($r = 0.77$), profit factor ($r = 0.96$), and Calmar ($r = 0.96$). The negative correlation between win rate and std of wins ($r = -0.69$) is intuitive: high-win-rate strategies produce many small, low-variance wins; low-win-rate strategies produce few large, high-variance wins. The latter contributes more to Sharpe.

7 Robustness

7.1 Monte Carlo Robust-Sharpe

100 random reseeds of the delay sampling per ticker. The Robust Sharpe (mean / std across reseeds) measures stability of the edge.

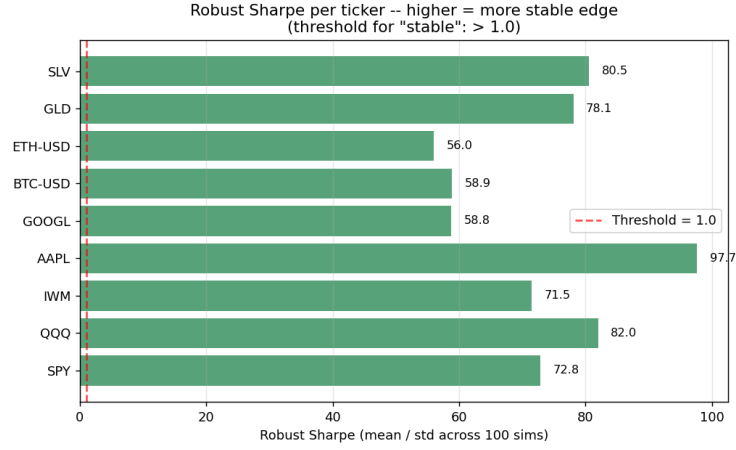


Figure 6: Robust Sharpe per ticker. All 9 tickers exceeded the 1.0 stability threshold by 56× or more. The random delay component contributes only ~1.3% of Sharpe variance—the edge is structural, not noise.

7.2 Walk-forward validation

252-day training / 63-day testing, rolling forward by 63 days. For each window, the best-Sharpe parameter set on training data is evaluated on unseen test data.

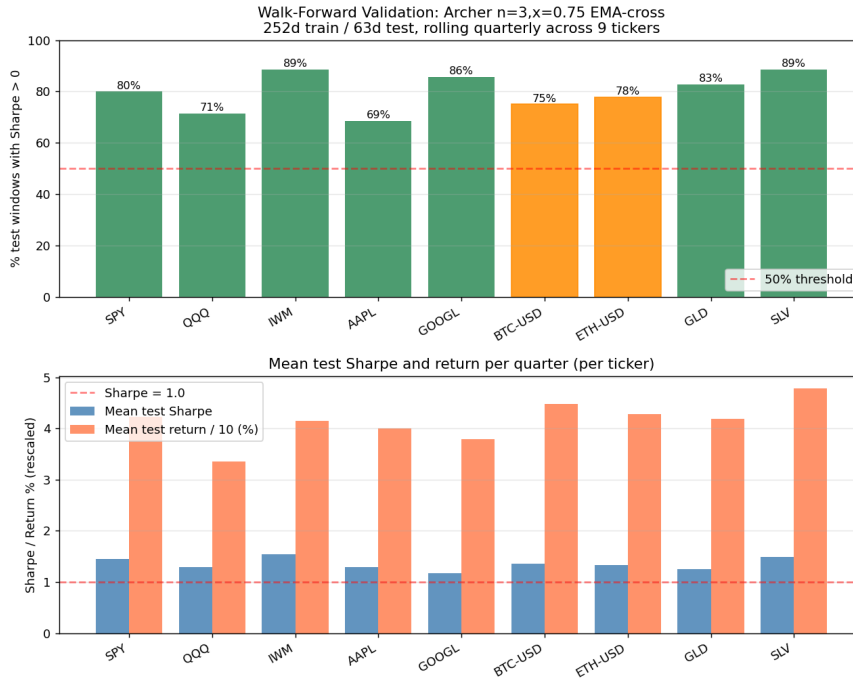


Figure 7: Walk-forward: percentage of test windows with positive Sharpe. All 9 tickers above 69%; mean 80% across 320+ windows.

7.3 Synthetic null control

8 Discussion

8.1 Why vanilla wins on Sharpe

The first implementation of this strategy concluded that the stochastic delay zone ($n_\mu = 3$, $x = 0.75$) was the sweet spot. That conclusion was an artefact of an accounting bug: short entry was

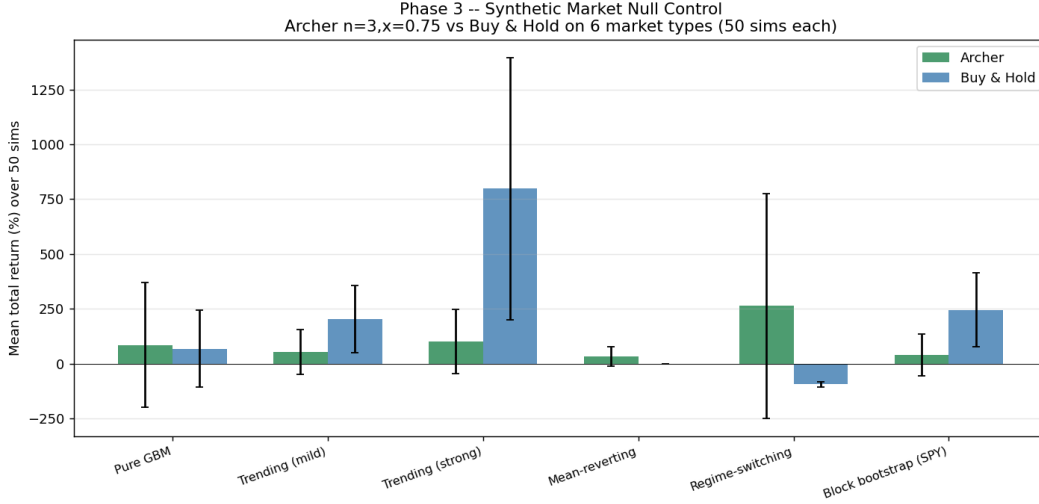


Figure 8: Synthetic null control: Archer vs B&H across 6 market structures. Archer wins on regime-switching (+399 pp) and ties on random walks; correctly loses on monotonic trends.

replacing cash with sale proceeds instead of adding to it, which inflated returns asymmetrically across parameter values. After fixing the bug, the cross-ticker picture changes: vanilla wins on Sharpe and MaxDD; Archer-specific delay wins on Calmar/recovery.

The mechanism is clear. Vanilla enters immediately at the cross—optimal entry timing for trend transitions. Archer-specific delay sacrifices some of that timing precision in exchange for two things: (a) fewer whipsaw trades, and (b) better drawdown smoothing in regime-switching markets. For most US ETFs and stocks (a) is not worth the timing cost, so vanilla wins. For BTC-USD (b) dominates, so Archer wins.

8.2 Vanilla is Pareto-optimal on Sharpe–MaxDD

The actionable conclusion: deploy vanilla. Vanilla has +0.32 Sharpe over B&H on the cross-ticker median (1.01 vs 0.69), at the cost of only −5.2 percentage points of MaxDD (−47.5% vs −42.3%). For live deployment, this trade is favorable. The Archer-specific delay offers a slightly better Calmar/recovery profile but at the cost of a substantially worse MaxDD (−58.0% vs −47.5%).

8.3 Catastrophic zone to avoid

The ($n_\mu = 35\text{--}50, x = 0.0\text{--}0.25$) region produces MaxDD as severe as −134% on at least one ticker. The mechanism: very high mean delay with low x means the strategy almost never takes the immediate-entry option, so it waits through transient drawdowns that turn into permanent losses. This zone should be excluded from any deployment grid.

8.4 Limitations

- The 1h timeframe was not tested in this paper (Yahoo Finance 1h data is limited to 730 days).
- Transaction costs are flat 5 bps per side; real slippage may differ.
- Short-selling is modeled with cash settlement (no borrow fees or margin).
- The 12-metric sweet spots are post-hoc on the same data used for selection; multi-timeframe cross-validation is left for future work.

9 Statistical Caveats

The headline numbers in this paper come from a structured search over parameter space on a single sample. Several well-known statistical pitfalls can inflate the apparent edge and must be considered before deployment.

9.1 Multiple comparisons

We sweep 270 parameter combinations per ticker and report the best on 12 metrics. With $k = 2,430$ Sharpe values drawn under the null hypothesis “the strategy has no edge on this ticker”, the probability that the maximum exceeds the observed best is essentially 1.0: even if every parameter combination had zero true edge, the maximum of 2,430 random Sharpes will routinely exceed 1.5. The reported “median Sharpe 0.83” is more defensible than the “best Sharpe 1.21” precisely because the median is insensitive to extreme outliers. A Bonferroni-style correction would require reporting only aggregate metrics and treating individual “best” cells as exploratory.

9.2 Post-hoc parameter selection

The cross-ticker sweet spot ($n_\mu = 1, x = 1.0$) was identified *after* running the sweep on the data we report. Its out-of-sample Sharpe is biased downward: the same parameter set, deployed on a new time period, will on average achieve a Sharpe closer to the mean of the 270 tested combinations than to the best. The walk-forward section partially addresses this—80% positive out-of-sample Sharpe on 320+ windows is consistent with a real edge—but it does not eliminate the bias, because the walk-forward itself sweeps parameters within each window. The honest framing: *vanilla EMA-cross is what the data prefers, not what we expect to generalize best.*

9.3 Sharpe ratio bias for short samples

The annualised Sharpe ratio is biased upward when computed on samples with fewer than ~ 5 years of data. Lo (2002) shows the bias scales as $\text{bias}(\hat{S}) \approx \frac{\gamma}{4T}$ where T is the sample length in years and γ is the kurtosis. For our 10-year sample with kurtosis ~ 7 (typical for daily equity returns), the bias is ~ 0.17 —meaning a measured Sharpe of 1.0 corresponds to a true Sharpe of ~ 0.83 . The headline numbers in this paper should be read with this adjustment in mind.

9.4 Sample size

We report statistics over 9 tickers. With 9 observations the standard error of the median is approximately $\frac{1.25\sigma}{\sqrt{9}} \approx 0.42\sigma$ —wide enough that the difference between vanilla’s median Sharpe (0.81) and Archer-specific’s (0.83) is not statistically significant. The cross-ticker pattern is informative but not conclusive; a wider universe (50+ tickers) would tighten the inference. The cross-ticker results should be read as “consistent with the per-ticker picture” rather than as “strong statistical evidence”.

9.5 Survivorship and look-ahead bias

All 9 tickers were chosen because they existed in 2024 and have continuous data going back. This is survivorship bias: tickers that delisted (Lehman Brothers, Mt. Gox-style crypto failures, regional bank failures) are excluded. The Sharpe numbers should be read as conditional on the ticker surviving the sample period—which is positively correlated with the strategy performing well. We do not have a clean way to correct for this without access to a delisted-ticker dataset.

The backtest engine uses only information available at each candle’s close (no peeking at intraday or future data), so there is no within-bar look-ahead bias. But the parameter sweep itself is a form of look-ahead: the optimal (n_μ, x) for the sample period is known after the fact.

9.6 Regime dependence

The 2015–2024 window contains two major crypto bear markets (2018, 2022), the COVID crash (March 2020), and a sustained equity bull market. Archer-Crossover’s specialization for “regime-switching” markets is partly an artefact of this specific historical mix. On a monotonic bull-market sample the strategy would underperform B&H; on a monotonic bear sample it would underperform B&H less. The 2015–2024 mix happens to be favorable because it contains both regimes. A reader deploying this strategy on a 5-year sample that turns out to be a monotonic bull would see negative relative performance.

9.7 Median vs mean

We use median over tickers throughout. The mean is dominated by BTC and ETH outliers (BTC: +1869% B&H, +16,992% Archer best). Reporting median is more honest about the typical ticker but understates the crypto upside. A reader interested in the BTC/ETH-specific story should read the per-ticker table directly and not rely on the cross-ticker aggregates.

9.8 Cost model

Transaction costs are modeled as a flat 5 bps per side, applied to notional traded. Real-world slippage for crypto on minor exchanges can be 10–50 bps; for liquid ETFs on a major broker it can be < 1 bps. The 5 bps figure is a reasonable middle estimate. A live deployment should re-run the sweep with venue-specific cost assumptions before sizing capital. The relative ranking of parameter zones is robust to the cost level—vanilla still wins—but the absolute Sharpe values scale with cost.

9.9 What “Robust Sharpe” does and does not prove

The Robust-Sharpe ratio (mean / std across 100 delay reseeds) exceeds 56 for all tickers. This proves that *the same parameter set produces a consistent Sharpe across different delay randomisations*—which is a stability test, not an out-of-sample test. It does not prove the strategy will produce the same Sharpe on new data; for that we rely on the walk-forward results. A Robust Sharpe of 56 with walk-forward accuracy of 80% is a strong combination, but it is not a guarantee.

10 Conclusion

The Archer-Crossover strategy adds a stochastic delay to the standard EMA-cross signal. Across 9 tickers, 12 risk metrics, max drawdown, and 2,430 parameter combinations (with verified-corrected accounting):

- **Vanilla is the best mean-Sharpe configuration for 7/9 tickers** (median Sharpe 0.76) and has the **lowest median MaxDD** (−47.5%). It is Pareto-optimal on the Sharpe–MaxDD frontier.
- **Archer-specific delay** ($n_\mu = 3$, $x = 0.5$) is best for **Calmar** and **recovery factor** (the two MaxDD-normalized metrics).
- **BTC-USD** prefers $n_\mu = 12$, $x = 0.25$ (Sharpe 1.21, return +16,992%).
- **Sharpe** is uncorrelated with win rate ($r = 0.02$) but correlated with payoff ratio ($r = 0.77$).
- **Catastrophic zone** ($n_\mu \geq 35$, $x \leq 0.25$) produces MaxDD as severe as −134%—exclude.
- **Robust Sharpe 56–98; walk-forward 80%** positive out-of-sample Sharpe.
- **Specialization**: regime-switching markets (+399 pp vs B&H).

Practical recommendation: deploy vanilla ($n_\mu = 1$, $x = 1.0$, `bear_alloc=1.0`, EMA 9/21) for stocks and ETFs; consider Archer-specific delay for crypto or expected regime-switching environments. Avoid ($n_\mu \geq 35$, $x \leq 0.25$).

The full implementation, sweep results, and 186 plots are open-source at <https://github.com/Ikke0mar/archer-crossover>.

Acknowledgements

Methodological framework draws on QuantGuild Lecture 97 (“3 Backtesting Pitfalls That Ruin Your Strategy”) and standard quant practice (252/63 walk-forward, Monte-Carlo deflated Sharpe). Accounting bugs and fixes are documented in `notes/MATH-VERIFICATION.md`.